CIE-III

Date: 28-1-25

Sub: TOC

Sub Code: CS354TA

Scheme and Solution

PART-A

1. T.M. for 2's compliment — 2 Marks
2. $\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, S\}$ — 1 Mark
3. At least 2 difference to be mentioned — 2 Marks
4. TM for equivalent to reg Exp $(0+1)^* 001 (0+1)^*$ — 2 Marks
5. $L = \{aa\}$ — 2 Marks
6. For every $\delta(q, a) = p$ of DFA write $\delta(q, a) = (p, a, R)$ in TM — 1 Mark.

PART-B

1. TM Definition — 1
   Language of TM as a language accepter — 1
   Language of TM as a Transducer — 1
   TM Construction for $L = \{ww^R \mid w \in \{a,b,c\}^*\}$ — 5
   TM trace for babcbab. — 2

2.a) LBA definition — 1
Language of LBA — 1
LBA for $L = \{ w \mid N_a(w) = N_b(w) \}$ — 2
Trace of LBA for babaab — 2

2.b) Note on chomsky heirarchy — 4

3.a) TM for string concatenation — 4
Trace for the given input — 2

3.b) Note on Multitape TM — 2
Equivalence of Multitape & Single tape TM — 2

4.a) PCP definition — 2
Solution to the given PCP — 4

4.b) proof for (i) — 2
proof of (ii) — 2

5.b) T.M for the string comparison for all 3 cases — 2+2+2 → 6

5.a) URG definition — 2
URG for the given language. — 2

~ o ~